

Program – 9: Organize Kubernetes Pods Using Labels and Namespaces

Project Structure

Program9/

```
├── namespaces.yaml
└── pods.yaml
```

Step 1: Create namespaces.yaml

nano namespaces.yaml

```
GNU nano 7.2 namespaces.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

Step 2: Create pods.yaml

nano pods.yaml

```
GNU nano 7.2 pods.yaml *
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
  env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
  env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
  env: prod
spec:
  containers:
  - name: frontend
```

Step 3: Apply Namespace YAML

kubectl apply -f namespaces.yaml

Step 4: Apply Pod YAML

kubectl apply -f pods.yaml

Step 5: Verify Pods by Namespace

- **Dev namespace**

kubectl get pods -n dev

- **Prod namespace**

kubectl get pods -n prod

Step 6: Check Pod Status

kubectl get pods

Step 7: Filter Pods Using Labels

- List frontend pods in dev

kubectl get pods -n dev -l app=frontend

- List backend pods in prod

kubectl get pods -n prod -l app=backend

```
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/dev0ps/program9$ nano namespaces.yaml
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/dev0ps/program9$ nano pods.yaml
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/dev0ps/program9$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/dev0ps/program9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/dev0ps/program9$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           35s
frontend-pod   1/1     Running   0           35s
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/dev0ps/program9$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           48s
frontend-pod   1/1     Running   0           48s
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/dev0ps/program9$ kubectl get pod
NAME          READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-6qrv2  1/1     Running   0           44m
nginx-deployment-6f9664446b-9ckzd   1/1     Running   0           44m
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/dev0ps/program9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod   1/1     Running   0           94s
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/dev0ps/program9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0          104s
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/dev0ps/program9$ █
```